



Mina Mikail

Mechatronics Engineer | CNC Automation & Manufacturing Software

Industrial Systems · Applied AI

I build the software that runs CNC manufacturing machines, and operate, integrate and maintain the machines myself.

CONTACT

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LinkedIn: linkedin.com/in/minamikailpk

Nationality: Swiss (CH + EU/EFTA)

Available: immediately · remote/relocation

CORE COMPETENCIES

Manufacturing & CNC software · Python automation · DXF / ISO & postprocessing · OPC-UA & machine integration · CAD/CAM (Rhino, Grasshopper, Revit) · Applied AI / LLMs · CNC operation, maintenance & tooling

LANGUAGES

English – C2

Arabic – C1

German – B1

Italian – A2

SUMMARY

Mechatronics engineer who runs CNC machines and builds the software that keeps them running. As the sole engineer for production systems at a four-person manufacturing startup, I operated, maintained and repaired the 3-/5-axis Biesse routers and developed the validation, machine-integration (OPC-UA) and automation tooling around them. Strong across CAD/geometry, manufacturing software and machine connectivity, backed by a public portfolio of tested engineering software.

EXPERIENCE

Production Development Engineer (Automation & CNC)

09/2024 – 02/2026

DataB GmbH (four-person industrial-automation startup) · Biedermannsdorf, Austria

- **Machine operation, maintenance & repair:** set up, operated and programmed the 3-/5-axis Biesse Rover routers and kept them running, repairing them on site and on call to restore production, with remote diagnosis comparable to the official Biesse partner's service.
- **Tooling & operator training:** ran the full tool lifecycle and inventory across all routers, procurement, sharpening logistics and tool optimization (e.g. moving film-faced material from a 10 mm to a 12 mm cutter, halving its passes and eliminating burn-through); wrote maintenance protocols and trained operators.
- **Validation & optimization pipeline:** built the production gate that checks every job before machining (geometry, collisions, sheet/material limits, cut-depth, missing toolpaths, material tagging, nesting efficiency) and optimizes toolpaths, after faulty files caused crashes and scrap. Across all 5 routers; ~300 modules, 33 test suites, CI.
- **Machine data & reverse-engineering:** with no documented way to export live data, reverse-engineered the controller interface and built an OPC-UA client (prototype to production Windows service) streaming to a database and Airtable; documented 328+ data files in a ~5,000-line reference, plus a real-time tool-change monitor.
- **CNC programming & postprocessing:** redesigned the NChops macro system to select tooling from the DXF layer name and automated batch-feeding of jobs; began an in-house DXF→ISO postprocessor to replace the NChops licenses (€6k per 3-axis, €10k per 5-axis machine).
- **Milling-time prediction (ML):** trained a Random-Forest milling-time predictor (42-feature model) on live production data, for quoting and scheduling.
- **Invoicing, reconciliation & reporting:** automated machining invoicing end to end (Excel→Airtable), separating billable from setup/test jobs; after finding that connection drops were silently losing ~10 jobs per machine per month (on a ~45-job/day schedule), built a second system that reconciles invoicing against the machines' own logs so no job is ever missed. Added automated project reports (DXF→PNG→DOCX) and cross-machine productivity analytics.
- **Software engineering:** consolidated ~118 ad-hoc scripts into one tested Python package (CI, packaged as Windows services); wrote wrappers for the vendor 32-bit DLL control APIs; co-built the in-house Grasshopper 3D→parts→DXF tool (later a Rhino plugin).

Python, ezdxf, Shapely, pandas, NumPy, scikit-learn, pytest, FastAPI, OPC-UA, DLL/API integration, PyInstaller, Airtable, Grasshopper.

IT Dispatch Technician

09/2023 – 03/2024

BCIC Swiss · Switzerland

- **Field dispatch & on-site support:** dispatched to client sites, including remote locations, to install, set up, power up and restore equipment on site.
- **Diagnosis, repair & server recovery:** diagnosed and repaired hardware faults across servers, workstations and peripherals, and recovered downed servers to restore service.
- **Systems & networking:** worked on Linux/Unix and Windows systems and troubleshooted networks (connectivity, switches/routers, cabling).
- **Imaging & provisioning:** imaged and pre-installed laptops and workstations at scale via PXE boot for client companies.
- **Parts & ticketing (ERP):** procured and tracked replacement parts through an ERP system, and logged, triaged and resolved support tickets.

Windows, Linux/Unix, server administration, networking, hardware repair, imaging/PXE, ERP.

Freelance BIM / Revit Modeling

08/2023 – 09/2023

Hpro GmbH · Zürich, Switzerland

- **Medical-technology modeling:** modeled medical-technology devices in Autodesk Revit from client specifications for a medtech-planning firm.
- **Parametric Revit families:** built reusable parametric Revit families (types and parameters) for the device models.
- **Iteration & delivery:** iterated prototypes through review cycles to the client's feedback, delivering each milestone on schedule as a remote freelancer.

Autodesk Revit, Revit families, BIM.

Engineering Intern (BIM)

2022 – 2023

Kreckox GmbH · Zürich, Switzerland

- **CAD & geometry foundation:** supported senior BIM engineers on live Revit deliverables within an ISO 19650 workflow; built and refined parametric Revit family templates (LOD 200–400), the CAD-geometry grounding later applied to CNC and manufacturing software.
- **Mechanical modeling & simulation:** produced mechanical models in Autodesk Inventor and ran basic stress and airflow simulations in SolidWorks to support design checks.
- **Automation & IoT:** wrote Python to automate repetitive modeling and data-extraction tasks; assisted with Arduino-based IoT and sensor prototyping feeding digital-twin and building-monitoring concepts.

Revit, Autodesk Inventor, SolidWorks, Python, Arduino; ISO 19650 / BIM.



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SELECTED PROJECTS (INDEPENDENT & OPEN-SOURCE, 2025 – PRESENT;

GITHUB.COM/PAPAKOFTES)

- **OMIM:** manufacturing geometry analysis that turns CAD drawings (DXF) into a validated machining model (a Manufacturing Geometry Graph) with design-for-manufacturing checks and feature detection. *Python, ezdxf, NetworkX, Shapely, FastAPI; tested, CI. Public.*
- **cnc-dxf-toolkit:** configurable DXF-prep library + CLI for CNC routing: cut start-point/lead-in, contour ordering, per-layer cut depth, layer normalization, nesting utilization and a pre-flight validator (geometry, bounds, machinability rules). *Python, ezdxf; 35 tests, CI. Public.*
- **cnc-cycle-time:** machine-learning cycle-time predictor: trains on a folder of DXFs plus a telemetry duration CSV to estimate the cut time of new parts. *Python, scikit-learn, ezdxf; tested, CI. Public.*
- **BiessePost:** a Python postprocessor for Biesse machines (DXF→ISO/BPP/CIX), built to the ISO 6983 standard: correct 5-axis kinematics with hard, non-overridable safety limits. *4,000+ automated tests, CI. Commercial product in active development (private repo); architecture available for technical discussion in interview.*
- **Interlock:** a fabrication engine that turns panel designs into joints (finger, dado, dovetail, mortise-and-tenon), nesting layouts and cut files with structural reports. *Python, ezdxf, Shapely, FastAPI. Commercial product in active development (private repo); architecture available for technical discussion in interview.*
- **Layla:** a private, on-device AI assistant that runs local language models with approval-gated tools and retrieval memory, fully offline. *Python, FastAPI, llama.cpp; 200+ tests, CI. Public.*
- **AMS-JobAssist:** an offline, privacy-first CV builder with a trainer dashboard; local AI, a 12-language interface and a full DPIA. *Python, FastAPI, SQLite; 662 tests, CI. Public.*
- **Hackathons & deployment:** Zero One Hackathon (Industrial-AI track, Infineon), top contributor, built the verification harness; Start Hackathon (Techbold track), built the safety and execution core of 'Sphinx', a technician-controlled AI troubleshooting copilot; deployed a synthetic-biology installation (Docker, Raspberry Pi) at the University:Future Festival 2026.
- **Personal fabrication:** designed and CNC-built enclosures and furniture end to end: 3D design, milling files, machine setup, operation and assembly.

SKILLS

Programming & Automation: Python (advanced), FastAPI, REST APIs, pandas, NumPy, ezdxf, Shapely, scikit-learn, pytest (unit & integration testing), CI (GitHub Actions), Docker, Git; SQL (SQLite, PostgreSQL); OPC-UA, DLL/API wrapping; Arduino C.

CNC & Machine Integration: 3-/5-axis Biesse Rover operation, programming, setup, maintenance & repair; DXF→ISO/G-code; NChops & layer-driven macros; postprocessing; machine reverse-engineering & telemetry; tooling & inventory management.

CAD & Computational Design: Rhino3D (advanced) & Grasshopper (parametric/visual programming); Revit (BIM & family modeling); SolidWorks/Inventor (basics); programmatic CAD (ezdxf, CadQuery).

Applied AI: Local LLMs (Ollama, llama.cpp), RAG & embeddings, agent/tool design, applied ML on production data.

DevOps & Data: Docker, Raspberry Pi, Caddy, systemd, Cloudflare Access, CI; real-time data pipelines (CSV, PostgreSQL, Airtable), reporting & reconciliation.

Embedded & Robotics: Arduino, ESP32, microcontrollers; soldering, perfboard & PCB etching; Fritzing/Tinkercad (above-basic).

EDUCATION

Mechatronics Engineering, October 6 University, Egypt · 2017 – 2021. Completed 4 of 5 years before relocating internationally (degree not completed).

Robotics/embedded: gyroscope-controlled RC helicopter; RFID access-control system; RC car; motor-driver, RF and Bluetooth circuits; from-scratch H-bridge charging circuit.

REFERENCES

On request: employment reference (DataB GmbH), project reference (Hpro GmbH, rated 'sehr gut'), plus additional industry-partner and client references.